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Simultaneous Lightwave Information and Power Transfer with Non-orthogonal Multiple Access

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***Abstract*—This paper proposes a non-orthogonal multiple access (NOMA) based simultaneous lightwave information and power transfer (SLIPT) scheme for downlink visible light communication (VLC) systems. A downlink VLC system with a light emitting diode (LED) transmitter and two users with photo detector (PD) receivers is considered. Also, a receiver circuit with a mode selector is proposed to be employed at each user. The proposed receiver circuit has energy harvesting capability by employing the power splitting (PS) as well as time switching (TS). The near user (NU) harvest energy utilizing PS protocol, whereas the far user (FU) utilized the TS protocol for energy harvesting. The numerical results are presented to illustrate that the user's data rates and the energy efficiency (EE) of the NOMA based SLIPT system show superior performance compared to the OMA based SLIPT system. Also, the harvested energy of the proposed energy harvesting scheme is within the reach of OMA based SLIPT systems.**

***Index Terms*—Energy efficiency, non-orthogonal multiple access, simultaneous lightwave information and power transfer and visible light communication.**

I. Introduction

The increasing demand for the wireless data service with higher rates has triggered the necessity for novel communication technologies that meets the requirement. They should be able to provide higher spectral efficiency (SE) while providing a higher energy efficiency (EE) as compared to the existing schemes. Visible light communication (VLC) has emerged as a cost effective communication technology which provides higher data rates. As VLC systems are using existing infrastructures of the illumination, these systems can provide acceptable EE while providing a higher SE. To further improve the EE of VLC systems, resources and power allocation optimization, simultaneous lightwave information and power transfer (SLIPT) and energy harvesting can be utilized [1], [2].

In SLIPT, optical signals are used for the simultaneous information and power transfer. Although, several research works have been conducted in RF based simultaneous wireless information and power transfer (SWIPT) [3], there are a few works reported in the literature on SLIPT. In [4], a dual-hop data communication system is proposed which consists of both VLC and RF links. This system has an energy harvesting relay which is capable of harvesting the energy from the VLC link. Also, the sum rate maximization problem was investigated for a downlink VLC system with SWIPT and a less complex algorithm was proposed to obtain the optimal solutions [5]. Moreover, the achievable rate region of a dual-hop VLC/RF system with optical energy harvesting relay is optimized in [6]. This hybrid VLC/RF communication system [4]- [6] is exploring an energy harvesting relay with a photo detector (PD) receiver. In [7], a SWIPT system for VLC which termed as SLIPT is presented. In this SLIPT system, a simple solar panel was used as the receiver and it is capable of both information receiving and energy harvesting. Although the electric power conversion of a solar panel is higher than an PD, detection sensitivity of an PD is higher than that of a solar cell [8]. Therefore, an PD as information receiver and a solar cell as an energy harvester is used in [8].

As non-orthogonal multiple access (NOMA) based VLC systems offers an enhanced SE, NOMA is a well suited multiple access method for VLC systems [9]. Specially, power domain NOMA provides an improved performance for VLC systems. A NOMA based hybrid VLC/RF communication system was considered in [10] to find the optimal grouping of the users. Also, in [11], a NOMA based VLC/RF cooperative communication system with SWIPT is proposed and the impact of the power splitting (PS) ratio for the system outage performance is studied. The LED emits incoherent light as per the driven current [12]. Therefore, the information to be transmitted is modulated into the instantaneous optical power of the LED. To keep the illumination level of an indoor VLC system, the brightness of the LED should be dimmed to a desired level. For a downlink VLC system, we propose a NOMA based SLIPT scheme. Also, we propose a receiver circuit, which has both PS and time switching (TS) capabilities according to the receiving mode i.e. near user (NU) or far user (FU). As per the author's knowledge, this is the first work incorporating the dual mode receiver in a NOMA based SLIPT system. Main contributions of this paper can be summarized as follows.

- A NOMA based SLIPT scheme is proposed for a downlink VLC system. Also, a dual mode receiver circuit is proposed which carries out the PS based energy harvest-

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ing at the NU, whereas TS based energy harvesting at the FU.

- The mathematical expressions for achievable rates and harvested energy of both NU and FU are derived. Also, an expression for the EE of the proposed SLIPT system is presented.
- The numerical results are presented to investigate the performance of the proposed SLIPT scheme. Achievable rates, harvested energy and the EE of the NOMA based SLIPT scheme is compared with those of an OMA based SLIPT scheme.

II. System Model

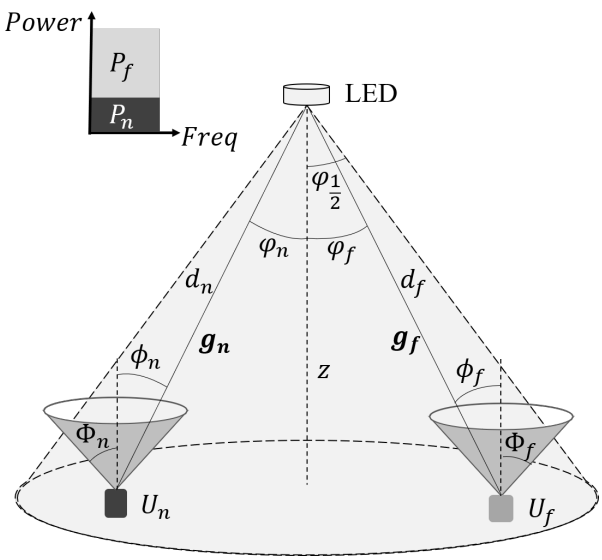

Fig. 1. A downlink VLC system with a LED transmitter and two users

This section presents a typical NOMA based downlink VLC system with SLIPT. The system consists of a light emitting diode (LED) as the transmitter and two users with the energy harvesting capabilities as depicted in Fig. 1.

A. SLIPT Transmitter

The block diagram of the NOMA based SLIPT transmitter is shown in Fig. 2.a. At the transmitter, data streams of two users are modulated and consequently, power domain NOMA is employed on the modulated signals. To obtain a non-negative transmission signal, a DC bias current, I_{DC} is added to the superposition mixture of two signals. The resulting NOMA coded signal, after adding the DC bias, can be expressed as

$$x(t) = \sqrt{P_s}(\alpha_n s_n + \alpha_f s_f) + I_{DC}, \tag{1}$$

where α_n and α_f are the power allocation coefficients of the NU and FU, respectively. Also, P_s is maximum power allocation of the NOMA and as per the NOMA principle $\alpha_n^2 + \alpha_f^2 = 1$. Moreover, s_n and s_f are the symbols to the NU and FU, respectively. After the optical drive, the transmitted optical signal can be expressed as

$$P_t = P_{LED}\left[\sqrt{P_s}(\alpha_n s_n + \alpha_f s_f) + I_{DC}\right], \tag{2}$$

where P_{LED} is the power of the LED per unit current in W/A. The average optical power can be given as

$$P_t^{avg} = P_{LED}(\sqrt{P_s} + I_{DC}) \tag{3}$$

The illumination control (dimming control) of the environment is performed by controlling the average optical power, thus controlling I_{DC} [8].

B. Channel model

Be assumed that the communication between the LED and an user is carried out only through a line-of-sight (LOS) path. The LOS direct current channel gain in VLC, g_i for the user U_i located inside the attocell of the LED is given by [13]

$$g_i = \begin{cases} \dfrac{A_i}{d_i^2} R_0(\varphi_i)\, T(\phi_i)\, g(\phi_i)\, \cos(\phi_i), & 0 \le \phi_i \le \Phi_i, \\ 0, & \phi_i > \Phi_i, \end{cases} \tag{4}$$

where $i \in \{n, f\}$. Also, A_i, Φ_i and ϕ_i, denote the PD area, the field of view (FoV) of the PD and the the angle of incidence of the U_i, respectively. d_i is the distance from the LED to U_i, φ_i is the angle of irradiation of the transmitter LED and $T(\phi_i)$ is the gain of the optical filter. The distance $d_i = z/\cos(\varphi_i)$, where z is the vertical distance between the transmitter and the receiver. Also, $R_0(\varphi_i)$ is the Lambertian radiant intensity of the LED transmitter and can be expressed as

$$R_0(\varphi_i) = \frac{m+1}{2\pi}\cos^m(\varphi_i), \tag{5}$$

where m represents the order of Lambertian emission, given as $m = -\ln(2)/\ln(\cos(\varphi_{1/2}))$, where $\varphi_{1/2}$ denotes the transmitter LED semi-angle at half power (half irradiation). In addition, $g(\phi_i)$ is the gain of the optical concentrator of the PD and it can be expressed as $g(\phi_i) = n^2/\sin^2(\Phi_i)$, where n is the refractive index.

C. SLIPT Receiver

The proposed receiver for NOMA based SLIPT systems is shown in Fig. 2.b. As per the distance from the LED transmitter, the mode selector selects the information decoding and energy harvesting mode. The user located near to the LED transmitter i.e. NU is assigned the PS mode whereas, the FU is assigned the TS mode.

III. Rate Analysis and Energy Harvesting

A. Achievable Rates of Users

As NU and FU are selected with two different modes, achievable rates and the harvested energy of them are different. The expressions of achievable rate and the harvested energy of the NU and FU are obtained as in following sub sections.

1) Achievable Rate of the NU: The received optical signal at the NU can be expressed as

$$P_n = g_n P_t = g_n P_{LED}\left[\sqrt{P_s}(\alpha_n s_n + \alpha_f s_f) + I_{DC}\right]. \tag{6}$$

The corresponding received electric signal at NU can be expressed as

$$y_n = \eta\, g_n P_{LED}\left[\sqrt{P_s}(\alpha_n s_n + \alpha_f s_f) + I_{DC}\right] + z_n, \tag{7}$$

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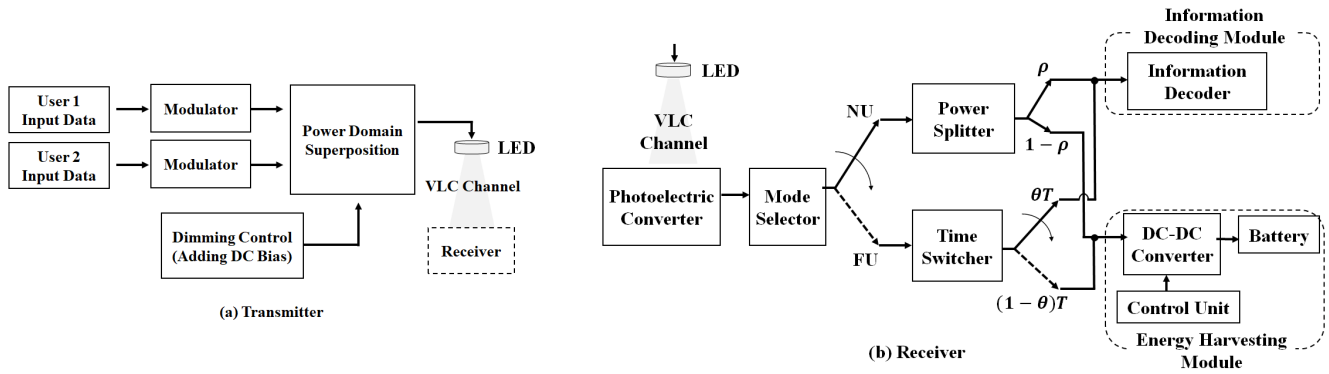

Fig. 2. A NOMA based SLIPT transmitter and receiver: a) transmitter; b) receiver with mode selector.

where z_n is the additive white Gaussian noise (AWGN) with zero mean and the variance σ_n^2 at the NU. Also, η is the responsivity of the PD in A/W. For the NU, PS mode is selected and power at the NU is split into two portions. The portion with power ratio ρ is used for information decoding and the portion with power ratio $1-\rho$ is used for energy harvesting. For the decoding of the received information, the DC component is removed and successive interference cancellation (SIC) is performed. Thus, the received signal-to-interference-plus-noise ratio (SINR) of the NU can be given as [14], [15]

$$\gamma_n = \frac{\rho\,\eta^2\,|g_n|^2\,\alpha_n^2\,P_s\,P_{LED}^2}{\sigma_n^2}. \tag{8}$$

The achievable rate of the NU can be expressed as

$$\begin{aligned} R_n &= \frac{B}{2}\log_2\left(1+\frac{e}{2\pi}\gamma_n\right) \\ &= \frac{B}{2}\log_2\left(1+\frac{e}{2\pi}\frac{\rho\,\eta^2\,|g_n|^2\,\alpha_n^2\,P_s\,P_{LED}^2}{\sigma_n^2}\right). \end{aligned} \tag{9}$$

2) Achievable Rate of the FU: The output signal of the PD at the FU can be expressed as

$$y_f = \eta\,g_f\,P_{LED}\left[\sqrt{P_s}(\alpha_n s_n + \alpha_f s_f) + I_{DC}\right] + z_f, \tag{10}$$

where z_f is the AWGN with zero mean and the variance σ_f^2 at the FU. At the FU, TS mode is selected and θT time duration is utilized for information decoding and $(1-\theta)T$ duration used for energy harvesting. After removing the DC component, the received signal is decoded. The received SINR of the FU can be given as

$$\gamma_f = \frac{\eta^2\,|g_f|^2\,\alpha_f^2\,P_s\,P_{LED}^2}{|g_f|^2\,\alpha_n^2\,P_s\,P_{LED}^2 + \sigma_f^2}. \tag{11}$$

The achievable rate of the FU can be expressed as [14], [15]

$$R_n = \theta T\frac{B}{2}\log_2\left(1+\frac{e}{2\pi}\frac{\eta^2\,|g_f|^2\,\alpha_f^2\,P_s\,P_{LED}^2}{|g_f|^2\,\alpha_n^2\,P_s\,P_{LED}^2 + \sigma_f^2}\right). \tag{12}$$

B. The Harvested Energy at Users

1) The Harvested Energy at NU: At the NU, $1-\rho$ portion of power is used for energy harvesting. It should be noted that the linear energy harvesting scheme is used. The harvested energy at the NU can be expressed as

$$P_{EH,n} = (1-\rho)\,T\,\eta^2\,|g_n|^2\,P_{LED}^2(P_s + I_{DC}^2). \tag{13}$$

2) The Harvested Energy at FU: At the FU, $(1-\theta)T$ duration is used for energy harvesting. The harvested energy at the FU can be expressed as

$$P_{EH,f} = (1-\theta)\,T\,\eta^2\,|g_f|^2\,P_{LED}^2(P_s + I_{DC}^2). \tag{14}$$

C. Energy Efficiency of the System

The total power consumption of the VLC system can be expressed as [16]

$$P_{con} = P_{LED}(\sqrt{P_s} + I_{DC}) + P_c, \tag{15}$$

where P_c is the power consumption of the circuit hardware. In SLIPT system, a portion of the transmitted energy is harvested at each user. Therefore, the total power consumption can be given as

$$P_{total} = [P_{LED}(\sqrt{P_s}+I_{DC}).+P_c]-(P_{EH,n}+P_{EH,f}). \tag{16}$$

The energy efficiency of a wireless communication system can be defined as $EE \triangleq R_{total}/P_{total}$. Therefore, the energy efficiency of the SLIPT enable VLC system can be given as

$$EE_{total} = \frac{R_n + R_f}{[P_{LED}(\sqrt{P_s} + I_{DC}) + P_c] - (P_{EH,n} + P_{EH,f})}. \tag{17}$$

IV. Results and Discussion

In this section, numerical results are presented to illustrate the performance of the NOMA based SLIPT system. To obtain the numerical results, we assume a NOMA based downlink VLC system in a room where the height of the room, z is 3 m. If not otherwise specified, parameters used to obtain the numerical results are given in the Table I.

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TABLE I
PARAMETERS USED FOR THE NUMERICAL RESULTS.

Parameter	Values
LED semi angle, $\varphi_{1/2}$	60°
FoV of the PD, Φ_i	50°
Responsivity of PD, η	0.4 A/W
Detection area of the PD, A_i	0.004 m^2
Refractive index, n	1.5
Gain of the optical filter, $T(\phi_i)$	1
Angle of incidence of PD, ϕ_i	$\phi_f = 45°$, $\phi_n = 50°$
Angle of irradiation of LED, φ_i	$\varphi_f = 40°$, $\varphi_n = 30°$
Power of the LED per unit current, I_{DC}	20 W/A
Bandwidth of the system, B	1 MHz

The NOMA based SLIPT system is compared with a OMA based SLIPT system. In the OMA based SLIPT system, frequency division multiple access (FDMA) is used as the multiple access technique and TS based energy harvesting is performed. TS based OMA SLIPT systems outperform the other energy harvesting techniques with respect to outage performance [17]. Therefore, in all comparisons, for the OMA based SLIPT, TS based SLIPT with $\theta = 0.5$ is used.

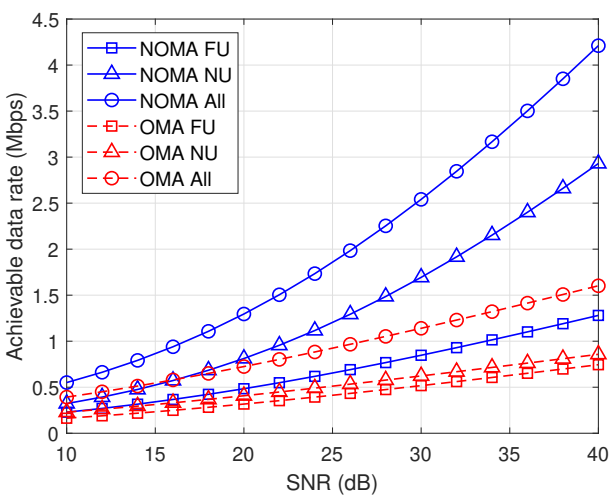

Fig. 3. Achievable rates vs. SNR of the NU and FU for NOMA and OMA based SLIPT systems over LOS channel.

Fig. 3 illustrates the comparison of achievable rates of the NU and FU. The ρ is set to 0.3 at the NU and $\theta = 0.3$ is used at the FU. For both OMA and NOMA SLIPT systems, T is set to 1 s. This comparison clearly illustrates that the NOMA based SLIPT system has higher data rates as both NOMA users utilized the total bandwidth of the system whereas, OMA users utilized a portion of the bandwidth. The rate comparison demonstrated that the data rates of the NOMA based SLIPT system are higher than the OMA based SLIPT system.

Fig. 4 illustrates the comparison of harvested energy with the DC bias for both OMA and NOMA based SLIPT systems. In this comparison, NOMA power allocation is set to 1 W and 4 W. Also, The ρ and θ is set to 0.3 at NU and FU. The harvested energy of all systems increases with the DC bias, I_{DC}. However, the harvested energy of the OMA based system

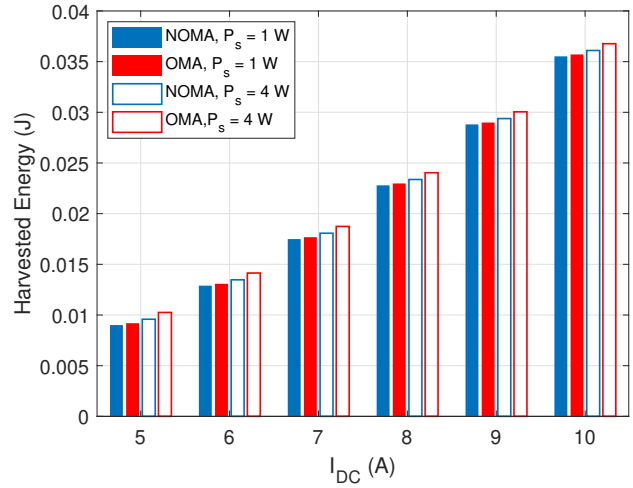

Fig. 4. Harvested energy of NOMA and OMA based SLIPT systems for different DC bias current, I_{DC} with $P_s = 1$ W and $P_s = 4$ W.

is higher than the NOMA based system. It is noteworthy that the difference between the harvested energy of OMA and NOMA based SLIPT systems is negligible. As per the distance from the transmitter, the NU and FU are selected with two different modes for energy harvesting. Therefore, the harvested at both users is increased for the proposed scheme.

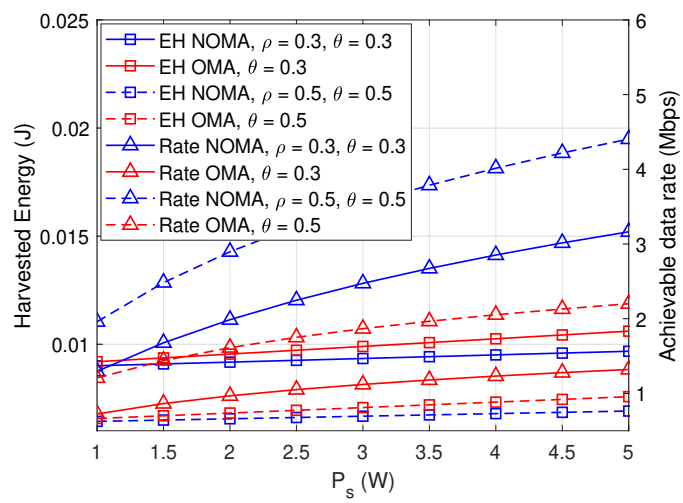

Fig. 5. Achievable sum rates and harvested energy of NOMA and OMA based SLIPT systems for different maximum power allocation of NOMA, P_s with $I_{DC} = 5$ A.

Fig. 5 demonstrates the comparison of the harvested energy and achievable sum rates with the maximum power allocation of NOMA, P_s at different values of ρ and θ. In this comparison, I_{DC} is set to 5 A. Both sum rates and the harvested energy is increased with P_s and similar rate and energy harvesting performances can be observed in comparison of OMA and NOMA SLIPT systems. Also, by allocating a higher power portion or time duration for the energy harvesting,

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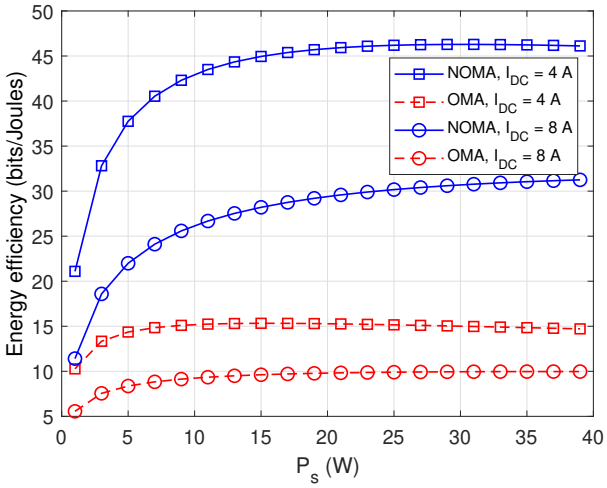

Fig. 6. EE of NOMA and OMA based SLIPT systems for different maximum power allocation of NOMA, P_s with $I_{DC} = 4$ A and $I_{DC} = 8$ A.

the harvested energy can be improved without a significant reduction of the rate performance.

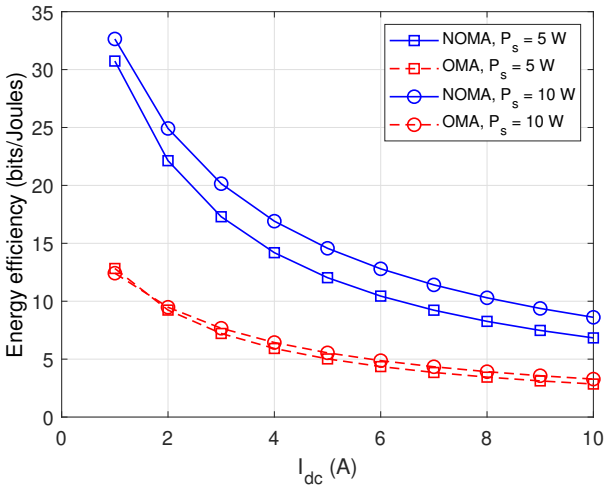

Fig. 7. EE of NOMA and OMA based SLIPT systems for different DC bias current, I_{DC} with $P_s = 5$ W and $P_s = 10$ W.

Fig. 6 depicts the comparison of the EE with the maximum power allocation of NOMA at different I_{DC} values for both OMA and NOMA based SLIPT systems. The selected I_{DC} values are 4 A and 8 A. The energy efficiency exhibits an increment nature with the maximum power allocation of NOMA and it saturates after a certain maximum power value. After the maximum value of P_s, when we increase the P_s, the increment of the power consumption is higher than the increment of total rate of the system. Therefore, EE shows saturation after a certain maximum power value and sometimes it shows a decrements (as depicted in OMA at $I_{DC} = 4$ A). Also, the increment of the DC bias value decreases the EE of both OMA and NOMA based SLIPT systems. The comparison of the EE with DC bias as shown in Fig. 7, it clearly demonstrates the decrements of the EE with I_{DC}. The harvested energy at different values of I_{DC} and P_s is given in Table II. In each EE comparison, EE performance of the NOMA based SLIPT system outperformed the OMA based system. The reason for the enhanced EE is that the NOMA based SLIPT systems harvest energy as near as OMA based systems even so, they provides higher data rates.

TABLE II
HARVESTED ENERGY OF ALL SYSTEMS WITH DIFFERENT VALUES OF I_{DC} AND P_s

I_{DC}	MA	Harvested energy (mJ)				
		$P_s = 1W$	$P_s = 2W$	$P_s = 3W$	$P_s = 4W$	$P_s = 5W$
2 A	NOMA	1.6	1.7	1.9	2.1	2.3
	OMA	1.8	2.1	2.5	2.8	3.2
4 A	NOMA	5.8	6.0	6.2	6.3	6.5
	OMA	6.0	6.4	6.7	7.1	7.4
6 A	NOMA	12.9	13.1	13.2	13.4	13.6
	OMA	13.1	13.4	13.8	14.1	14.5
8 A	NOMA	22.8	23.0	23.1	23.3	23.5
	OMA	23.0	23.3	23.7	24.0	24.4
10 A	NOMA	35.5	35.7	35.9	36.0	36.2
	OMA	35.7	36.1	36.4	36.8	37.1

V. CONCLUSIONS

A NOMA based SLIPT scheme has been proposed for a downlink VLC system with a LED transmitter and two users. A single receiver circuit with a mode selector has been proposed at both NU and FU. This receiver circuit has the capability of performing both PS and TS based SLIPT. For purpose of energy harvesting, the PS protocol has been utilized at the NU whereas, the TS protocol has been utilized at the FU. Numerical results are presented to illustrate the superiority of the performance of the NOMA based SLIPT system over a OMA based SLIPT system. User data rates, harvested energy and EE of the NOMA based systems have been compared with those of the OMA based SLIPT system. NOMA based systems outperform the OMA based system in rate performance as well as in EE. Although the harvested energy of the OMA based system is slightly higher than that of the NOMA based system, the EE performance is higher in NOMA based SLIPT systems. As the NOMA based SLIPT system illustrates a higher SE and EE, the PS and TS coefficients can be set to further improve the EE of these systems without compromising much of the SE.

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